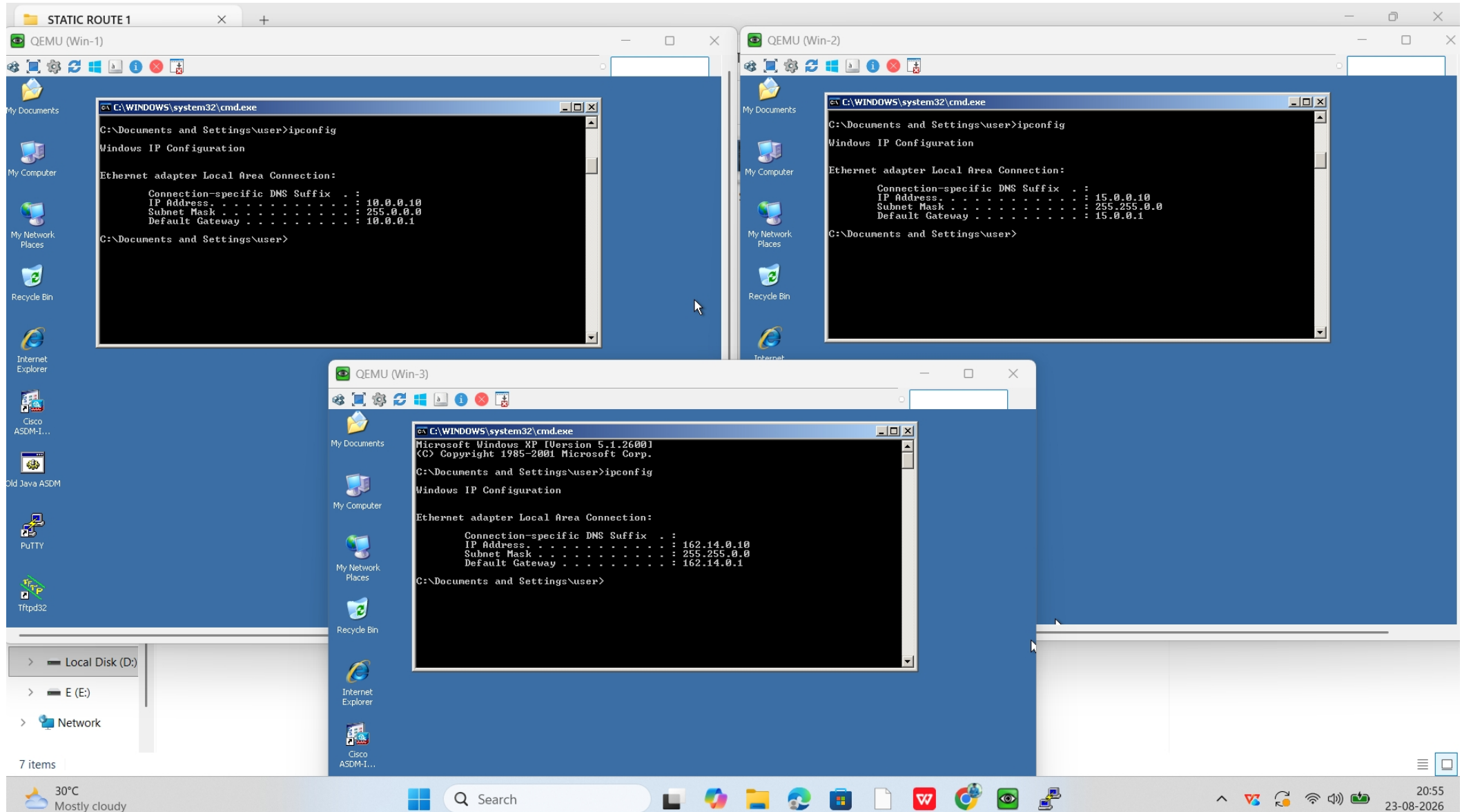


STATIC ROUTE PROJECT -1

ASSIGN THE IP ADDRESSES FOR ALL THE PC'S:-



ASSIGN THE IP ADDRESSES FOR ALL THE ROUTERS INTERFACES:-

The image displays a network configuration interface with four routers (R1, R2, R3, R4) and their interface IP addresses. The interface includes a network diagram and a taskbar at the bottom.

R1 Configuration:

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	10.0.0.1	YES	manual	up	up
Ethernet0/1	20.0.0.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

R2 Configuration:

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	20.0.0.2	YES	manual	up	up
Ethernet0/1	30.0.0.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

R3 Configuration:

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	30.0.0.2	YES	manual	up	up
Ethernet0/1	172.19.0.1	YES	manual	up	up
Ethernet0/2	21.0.0.1	YES	manual	up	up
Ethernet0/3	unassigned	YES	TFTP	down	down

R4 Configuration:

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	15.0.0.1	YES	manual	up	up
Ethernet0/1	172.19.0.2	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

Network Diagram:

The diagram shows four routers (R1, R2, R3, R4) connected in a mesh topology. R1 is connected to R2, R3, and R4. R2 is connected to R1 and R4. R3 is connected to R1 and R4. R4 is connected to R2 and R3. The IP addresses for the interfaces are: R1 (10.0.0.1, 20.0.0.1, unassigned, unassigned), R2 (20.0.0.2, 30.0.0.1, unassigned, unassigned), R3 (30.0.0.2, 172.19.0.1, 21.0.0.1, unassigned), and R4 (15.0.0.1, 172.19.0.2, unassigned, unassigned). The diagram also shows a host (Win-3) connected to R4 with IP address 162.14.0.10.

Taskbar:

The taskbar at the bottom shows the system clock (20:22, 23-08-2026), network status (32°C, Mostly cloudy), and various application icons (Search, File Explorer, Edge, Word, Chrome, etc.).

R5

% Type "show ?" for a list of subcommands

R5#sh ip int bri

Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	21.0.0.2	YES	manual	up	up
Ethernet0/1	192.168.168.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

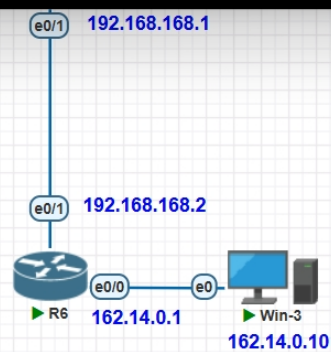
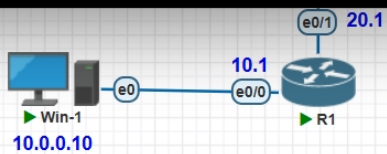
R5#

R6

R6#sh ip int br

Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	162.14.0.1	YES	manual	up	up
Ethernet0/1	192.168.168.2	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

R6#



Physical

01 : 00 : 00

START

32°C
Mostly cloudy



Search

20:27
23-08-2026

ADD A STATIC ROUTES FOR MISSING ROUTES IN ALL THE ROUTERS:-

Gateway of last resort is not set

```
S 10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.0.0.0/8 is directly connected, Ethernet0/0
L 10.0.0.1/32 is directly connected, Ethernet0/0
S 15.0.0.0/8 [1/0] via 20.0.0.2
C 20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 20.0.0.0/8 is directly connected, Ethernet0/1
L 20.0.0.1/32 is directly connected, Ethernet0/1
S 21.0.0.0/8 [1/0] via 20.0.0.2
S 30.0.0.0/8 [1/0] via 20.0.0.2
S 162.14.0.0/16 [1/0] via 20.0.0.2
S 172.19.0.0/16 [1/0] via 20.0.0.2
S 192.168.168.0/24 [1/0] via 20.0.0.2
R1#
```

Gateway of last resort is not set

```
+ - replicated route, % - next hop override, p - overrides from PfR
S 10.0.0.0/8 [1/0] via 20.0.0.1
S 15.0.0.0/8 [1/0] via 30.0.0.2
C 20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 20.0.0.0/8 is directly connected, Ethernet0/0
L 20.0.0.2/32 is directly connected, Ethernet0/0
S 21.0.0.0/8 [1/0] via 30.0.0.2
C 30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 30.0.0.0/8 is directly connected, Ethernet0/1
L 30.0.0.1/32 is directly connected, Ethernet0/1
S 162.14.0.0/16 [1/0] via 30.0.0.2
S 172.19.0.0/16 [1/0] via 30.0.0.2
S 192.168.168.0/24 [1/0] via 30.0.0.2
R2#
```

Gateway of last resort is not set

```
S 10.0.0.0/8 [1/0] via 30.0.0.1
S 15.0.0.0/8 [1/0] via 172.19.0.2
S 20.0.0.0/8 [1/0] via 30.0.0.1
C 21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 21.0.0.0/8 is directly connected, Ethernet0/2
L 21.0.0.1/32 is directly connected, Ethernet0/2
C 30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 30.0.0.0/8 is directly connected, Ethernet0/0
L 30.0.0.2/32 is directly connected, Ethernet0/0
S 162.14.0.0/16 [1/0] via 21.0.0.2
C 172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 172.19.0.0/16 is directly connected, Ethernet0/1
L 172.19.0.1/32 is directly connected, Ethernet0/1
S 192.168.168.0/24 [1/0] via 21.0.0.2
R3#
```

Gateway of last resort is not set

```
+ - replicated route, % - next hop override, p - overrides from PfR
S 10.0.0.0/8 [1/0] via 172.19.0.1
C 15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 15.0.0.0/8 is directly connected, Ethernet0/0
L 15.0.0.1/32 is directly connected, Ethernet0/0
S 20.0.0.0/8 [1/0] via 172.19.0.1
S 21.0.0.0/8 [1/0] via 172.19.0.1
S 30.0.0.0/8 [1/0] via 172.19.0.1
S 162.14.0.0/16 [1/0] via 172.19.0.1
C 172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 172.19.0.0/16 is directly connected, Ethernet0/1
L 172.19.0.2/32 is directly connected, Ethernet0/1
S 192.168.168.0/24 [1/0] via 172.19.0.1
R4#
```

Physical 01 : 00 : 00 START

32°C Mostly cloudy 20:32 23-08-2026

ALL PCS' IP ADDRESSES ARE REACHABLE THROUGH THE ROUTERS:-

The screenshot displays a PNETLab topology simulation environment. At the top, a browser window shows several tabs: PNETLab | Topology, ChatGPT, (19) WhatsApp, Notes/CCNA/SN-CA-B, CCNA-TEST/LAB PRACTICE, New Tab, and Ask Gemini. Below the browser, two QEMU (Win-1) and QEMU (Win-2) virtual machines are running. Both VMs show a Windows XP desktop with a command prompt open, displaying the output of the 'ipconfig' and 'ping' commands.

QEMU (Win-1) Command Prompt Output:

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\User>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 10.0.0.10
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 10.0.0.1

C:\Documents and Settings\User>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\User>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=2ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\User>
```

QEMU (Win-2) Command Prompt Output:

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\User>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 15.0.0.10
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 15.0.0.1

C:\Documents and Settings\User>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=3ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\User>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=3ms TTL=124
Reply from 162.14.0.10: bytes=32 time=3ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\User>
```

At the bottom of the screen, a network diagram shows a router (R6) connected to two PCs (Win-1 and Win-2). The router's interface e0/0 is labeled with IP address 162.14.0.1. The PCs are labeled with IP addresses 162.14.0.10 and 162.14.0.10. A status bar at the bottom indicates the simulation is running for 01:00:00.

← → ↻ labs.surennetwork.com/legacy/topology

☆ Paused

Notifications

My Documents

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM-1...

Old Java ASDM

Win-1
10.0.0.10

PutTY

Tftp32

QEMU (Win-3)

C:\WINDOWS\system32\cmd.exe

Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.
C:\Documents and Settings\User>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

 Connection-specific DNS Suffix . :
 IP Address : 162.14.0.10
 Subnet Mask : 255.255.0.0
 Default Gateway : 162.14.0.1

C:\Documents and Settings\User>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=4ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123

Ping statistics for 10.0.0.10:
 Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
 Approximate round trip times in milli-seconds:
 Minimum = 2ms, Maximum = 4ms, Average = 2ms

C:\Documents and Settings\User>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=1ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 15.0.0.10:
 Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
 Approximate round trip times in milli-seconds:
 Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\User>

Win-2
15.0.0.10

R6
162.14.0.1

Win-3
162.14.0.10

Physical

01 : 00 : 00 START

30°C
Mostly cloudy

Search

20:48
23-08-2026